

Vibhasri Kandukuri

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EDUCATION

Purdue University

B.S. in Computer Science, Machine Learning Concentration, Certificate in Applications of Data Science

West Lafayette, IN

Aug. 2023 – Dec. 2026

EXPERIENCE

• Team Lead and Software Engineering Researcher

Aug 2024 - May 2025

AgRPA, The Data Mine, Purdue University

- Engineered an autonomous drone mapping and targeted pesticide-spraying system that reduced chemical use and operational costs
- Developed a Python geospatial processing pipeline to convert shapefiles into optimized QGroundControl flight waypoints, improving battery utilization by 20% and increasing image resolution
- Led cross-functional coordination across AI, software, and mechanical engineering teams to integrate autonomous navigation, imaging, and spraying subsystems

• Machine Learning and Data Science Researcher

Aug 2023 - May 2024

Caterpillar, The Data Mine, Purdue University

- Built Flutter mobile application for EV fleet managers to monitor battery health and receive real-time system notifications
- Preprocessed and applied ARIMA forecasting on telemetry records to improve battery utilization and charging efficiency

• Scrum Master & Undergraduate Teaching Assistant

Aug 2025 - May 2026

Lincoln Financial Group (LFG) & Zimmer Biomet, The Data Mine, Purdue University

- Guided student teams collaborating with LFG to develop logistic regression and decision tree models analyzing annuity outcomes and financial risk factors
- Coordinated Zimmer Biomet student teams building healthcare data enrichment workflows using NPI Registry web scraping to map provider identifiers to enterprise hospital datasets and support Snowflake Cortex AI querying
- Facilitated Agile sprint planning, stakeholder communication, and technical reviews across 8-11 member cross-functional teams

• Data Science Intern

May 2024 - Aug 2024

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- Analyzed patient health records using Pandas and Matplotlib to identify correlations and risk factors associated with diabetes onset
- Performed data preprocessing, feature analysis, and exploratory analysis on structured healthcare datasets

PROJECTS

• Shell Interpreter

Nov 2024

Created a Unix shell in C featuring fork/exec-based process execution, piping, file redirection, subshell evaluation, environment and wildcard expansion, signal handling, zombie-process cleanup, and interactive command history using Lex/Yacc parsers.

• Personal Malloc in C

Jan 2025

Implemented a malloc/free allocator in C using segregated free lists, block splitting/coalescing, and sbrk()-based heap expansion to manage dynamic memory allocation.

• Simple C Compiler

Mar 2024

Built a compiler in C using Lex/Yacc to tokenize, parse, and generate x86-64 assembly for a reduced C language.

• Java Marketplace

Nov 2023

Built a multithreaded Java marketplace application with socket-based client/server communication, synchronized shared-state management, persistent data storage, and Swing-based buyer/seller interfaces.

• Android SQLite Java Bookstore

Apr 2023

Developed an Android application in Java with an SQLite backend supporting inventory management, order tracking, offline persistence, and search functionality.

SKILLS

Languages: C, Java, Python, Bash, SQL, R, Dart, HTML/CSS

Frameworks/Libraries: React, Pandas, NumPy, Scikit-Learn, SQLite

Systems & Concepts: Multithreading, Socket Programming, Systems Programming, Process Management, Memory Allocation, x86-64 Assembly, Lex/Yacc

Tools: Git, Linux/Unix, Jupyter, Agile Development, LaTeX